

Haytam El Ouarrat

Machine Learning & Data Science Engineer

Toulouse, France
+212 708 052 763
eo.haytam@gmail.com
haytam-el-ouarrat
haytamelouarrat00
Portfolio

Technical Skills

Languages Python, C++, SQL, MATLAB, C#
ML/AI PyTorch, TensorFlow, Scikit-learn, LightGBM, XGBoost, JAX
Libraries OpenCV, Polars, Pandas, NumPy, FastAPI, PCL, Optuna
Tools & DevOps Git, Docker, AWS, Azure, MLflow, Weights & Biases (W&B)

Experience

- Oct 2025 – **Research Assistant**, Mohammed VI Polytechnic University, Rabat, Morocco
Present
 - Developed a closed-loop visual servoing framework using **3D Gaussian Splatting** and **NeRFs** for high-precision, depth-sensorless camera navigation.
 - Implemented and benchmarked **Feature-Based IBVS** with **XFeat** and **MoGe-2** to recover metric scene geometry and validate trajectories.
 - Rigorously evaluated model performance across 350+ synchronized evaluation scenarios, achieving state-of-the-art **sub-millimeter precision** (452 μm RMSE) and robust spatial tracking capabilities.
- Mar 2025 – **Computer Vision Engineer Intern – R&D**, SteelSeries, Lille, France
Aug 2025
 - Designed and trained a Teacher-Student framework combining classical and neural methods to accelerate in-game event detection for the SteelSeries Moments software.
 - Explored and optimized lightweight Vision Transformers and CNNs, achieving a **12.4x speedup** over traditional SIFT feature matching while maintaining high accuracy.
 - Curated a custom dataset of 127k+ gaming footage samples utilizing synthetic transformations, achieving ~81% keypoint matching accuracy through model optimization.
- Jan 2024 – **Perception and Computer Vision Team Lead**, TLSe Racing, Toulouse, France
Feb 2025
 - Led a team of 7 engineers to develop a robust, real-time perception pipeline for a Formula Student Driverless vehicle.
 - Implemented real-time 3D object detection and tracking of track cones utilizing **YOLOv11** and custom inference optimization.
 - Engineered a sensor fusion algorithm integrating LiDAR, stereo cameras, and IMU data pipelines, improving spatial accuracy by 26%.

Selected Projects

- 2026 **AeroDrift: Predictive Maintenance API**
 - Built a production-grade Machine Learning system to predict Remaining Useful Life (RUL) of turbofan engines using high-frequency IoT telemetry data.
 - Developed a stateful **FastAPI** microservice processing continuous data streams with **Polars** and running a **LightGBM** model optimized with a NASA-specific Asymmetric Loss Function.
- 2025 **Curseless: AI Audio Profanity Censor**
 - Developed a robust deep learning system for automated detection and censorship of profanity in audio content.
 - Leveraged **OpenAI's Whisper** for precise speech recognition and implemented reinforcement learning techniques to continuously improve detection accuracy.
- 2026 **VISNAV: Visual Servoing Navigation Framework**
 - Created a research framework for camera navigation within photo-realistic 3D scenes (3DGS, NeRF) achieving sub-millimeter translational and sub-degree angular convergence.

Education

- Oct 2025 **Master of Engineering, Robotics and Smart Systems**, Engineering School of the University of Toulouse, Toulouse, France